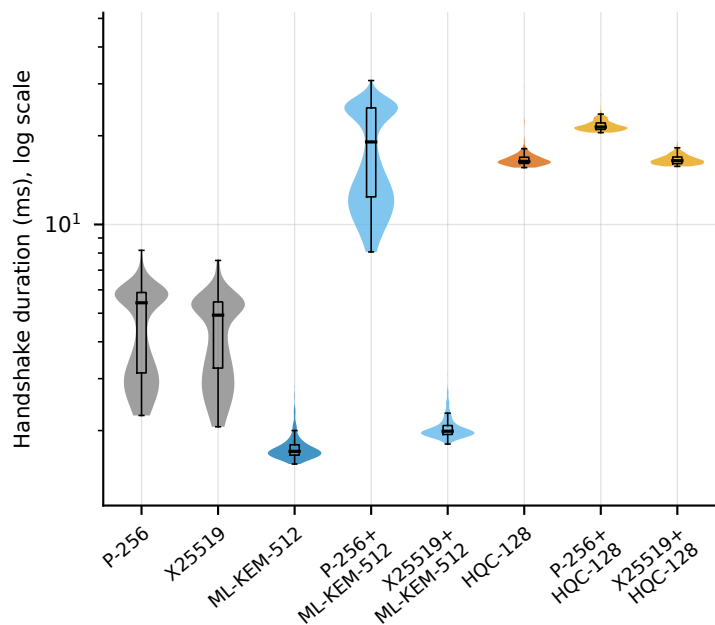
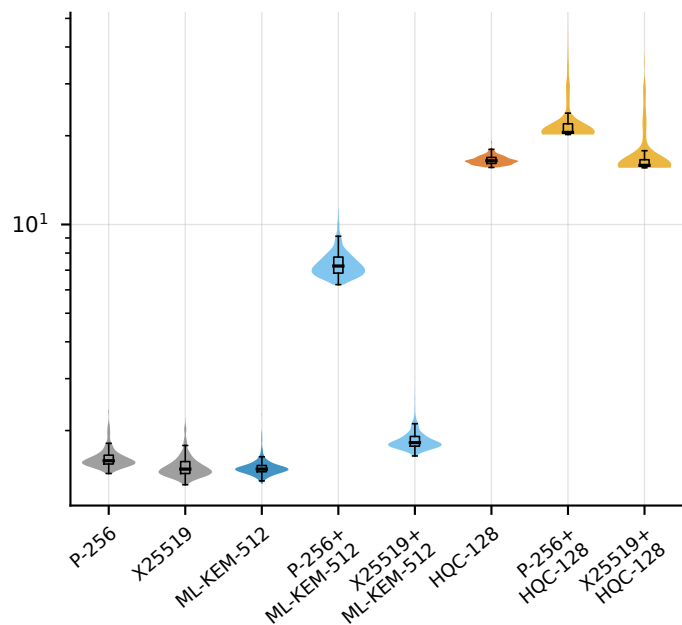


(a)

ed25519 (classical)

**(b)**

mlds44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mlds44 Mean (ms)	mlds44 CV (%)	mlds44 Max (ms)
P-256	4.72	30.6	8.17	1.62	8.5	2.36
X25519	4.49	29.2	7.55	1.52	9.1	2.20
ML-KEM-512	1.75	9.7	2.86	1.50	6.1	2.29
P-256+ML-KEM-512	18.7	33.2	30.8	7.41	10.8	11.3
X25519+ML-KEM-512	2.05	8.8	3.15	1.86	7.9	3.06
HQC-128	16.6	5.8	29.3	16.5	3.4	19.2
P-256+HQC-128	21.6	3.9	27.5	22.5	19.8	53.1
X25519+HQC-128	16.6	4.2	25.0	17.7	22.7	38.4